

End of year report

Plasma processing of PAHs and related molecules – Theoretical and experimental approaches

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1. Introduction

The EUROPAH European Training Network aims to study the extensive and ubiquitous role of Polycyclic Aromatic Hydrocarbons (PAHs) in space. In the context of the energetic processing of PAHs, the present PhD project focuses on their interaction with energetic electrons, i.e. plasma processing of naphthalene as a PAH model compound. The main objective is to identify fragmentation patterns under plasma conditions so as to understand reaction mechanisms for PAHs' decomposition in the interstellar medium (ISM), both from a theoretical and experimental approaches.

The present document reports the progress made in the period May 2018 – April 2019. The objectives set out in the future plans section of the End of Year report 2018 were accomplished as follows:

- The manuscript on plasma kinetic modelling of CO₂ dissociation reactions (already submitted by the time of the previous report) was finally published in the Journal of CO₂ utilization (see section 6 for more details).
- The mass spectrometric characterisation of a naphthalene low pressure radio frequency (LP-RF) plasma was completed. The results are reported in section **Error! Reference source not found.** and were published in the Journal of Physical Chemistry A (see section 6 for more details).
- R-matrix calculations are being performed in order to calculate electron impact cross sections as input parameters for the plasma kinetic model. The results obtained so far are presented in section 3.
- A plasma kinetic model for naphthalene decomposition in a dielectric barrier discharge (DBD) reactor is being developed. Preliminary results are presented in section 4.

Apart from the works reported here, DFT calculations are being carried out to characterise Nickel-based catalysts and their interactions with CO₂ molecules. This study is not reported here as it is not PAH-related.

2. Mass spectrometry experiments

The experimental set up is not described here given that it was introduced in the End of Year report 2018. Only a scheme is presented in Figure 1, showing two additions with respect to the 2018 set-up, i.e., the hot water bath that proved necessary to enhance vaporisation and the

function generator to pulse the plasma discharge as required for the negative ion detection. The rig reached a base pressure of $\sim 2.7 \times 10^{-3}$ Pa (2×10^{-5} Torr) before naphthalene vapours were introduced.

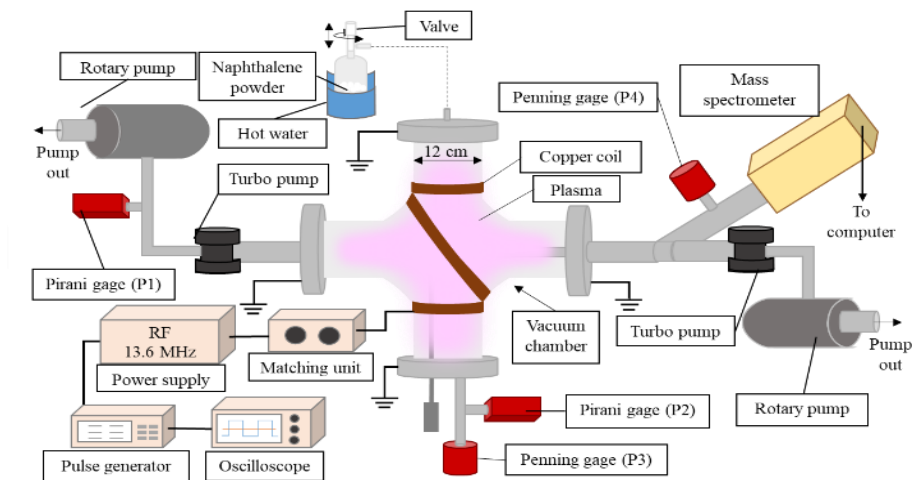


Fig. 1 – Schematic diagram of the experimental setup

2.1. Results – positive ions

Mass spectra of positive ions in the RF capacitively coupled plasma were collected at 0.67 Pa and several discharge power values in the range 10-95 W, using the SIMS mode of the mass spectrometer. Each spectrum was averaged over 3-4 scan cycles. Figure 2 shows the spectrum recorded at 95 W, featuring a strong peak at $m/z = 128$ amu that corresponds to the naphthalene radical cation ($C_{10}H_8^+$). The latter is the main cation in the naphthalene plasma as the signals of all the other ions detected are roughly an order of magnitude lower than that of $C_{10}H_8^+$. Hence, even at the highest power applied in this work, the fragmentation of the molecule is still incipient. Nonetheless, there is a correlation between the power applied and the degree of fragmentation, which is highlighted in Figure 3. The latter presents the spectra collected at different discharge power values ranging from 10 to 95 W. In Figure 3, each spectrum is normalised by the peak at $m/z = 128$ amu ($C_{10}H_8^+$), which allows for a direct comparison of the degree of fragmentation between different spectra.

The fragmentation patterns of naphthalene under plasma conditions are analysed based on the spectrum recorded at 95W, given that the qualitative results are analogous for every discharge power. Various peaks in Figure 2 are broadened both to the left and to the right by one or two

units, thus indicating addition or loss of hydrogen atoms. For instance, the peak at $m/z = 128$ amu appears to be broadened by peaks at $m/z = 127$ amu and $m/z = 126$ amu, which indicate the loss of one or two hydrogen atoms resulting in $C_{10}H_7^+$ and $C_{10}H_6^+$, respectively. This result is in agreement with the reported photodissociation channels [1–6] and also with a theoretical study on the chemistry of the naphthalene radical cation which indeed predicts loss of both atomic and molecular hydrogen [7].

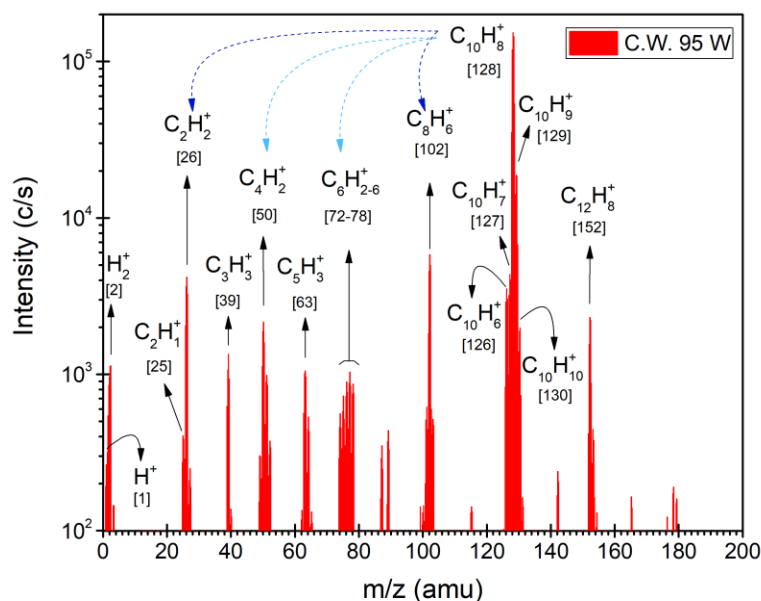


Fig. 2 – Positive ions mass spectrum at 95W highlighting the postulated fragmentation patterns giving rise to each signal.

The main naphthalene fragments are represented by the peaks at $m/z = 26$ amu and $m/z = 102$ amu, assigned to species with the formulae $C_2H_2^+$ and $C_8H_6^+$, respectively. This suggests that the major dissociation channel of the naphthalene radical cation involves the loss of an acetylene molecule (C_2H_2), giving rise to cation detected at $m/z = 102$ amu (see dark blue dashed arrows in Figure 2). The neutral acetylene molecule could be further ionised in the plasma by electron impact or charge exchange interactions and thus, appear in the spectrum at $m/z = 26$ amu. The fragmentation proposed agrees with the photodissociation related literature [1–6]. Moreover, this process has been theoretically described as involving several isomerisation routes of the naphthalene radical cation which lead to different isomers of the resulting $C_8H_6^+$ cation [7]. Besides, from an astrophysical perspective, it has also been reported that PAHs in a hot post-shock gas are destroyed by electron collisions via C_2H_n loss [8].

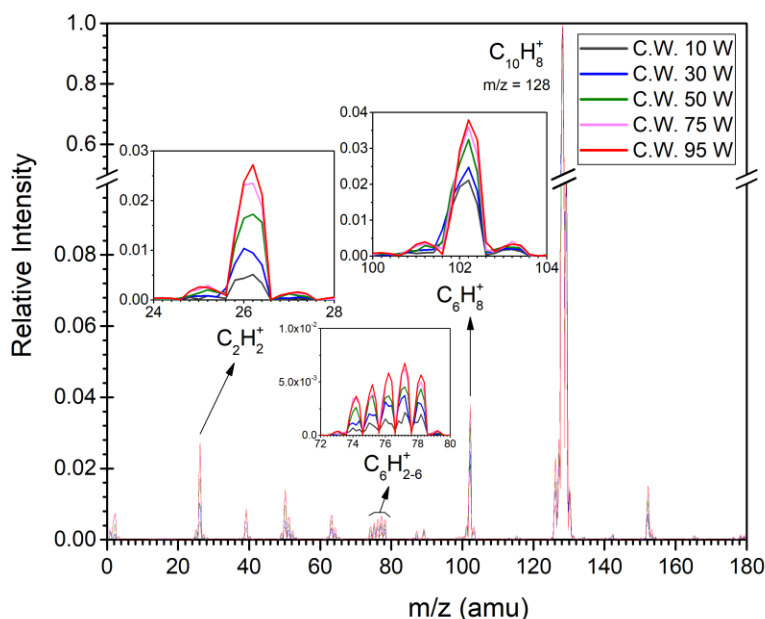


Fig. 3 – Spectra at different discharge power values in the range from 10 to 95 W highlighting its influence in the degree of naphthalene fragmentation.

Finally, the scan of positive ions (Figure 2) shows a signal at $m/z = 152$ amu, which is clearly not a naphthalene fragment as its mass is higher than 128 amu. This signal corresponds to a cation with molecular formula $C_{12}H_8^+$, which could be acenaphthalene or ethynylnaphthalene. However, there is no theoretical evidence to support a statement on relative stability of these two isomeric cations. Nonetheless, given the mass difference between this peak and that of naphthalene (24 amu), and considering the presence of $C_2H_2^+$ in the system, we postulate that the hydrogen abstraction – acetylene addition (HACA) mechanism is implied in the formation of $C_{12}H_8^+$. The HACA mechanism has been widely employed to rationalise the growth of PAHs in the ISM and circumstellar outflows of carbon stars [9–17]. Recently, some experimental evidences of this mechanism under combustion-like conditions or thermal activation have been reported [13,14,18,19]. To the best of our knowledge, the present study shows the first experimental evidence of a PAH molecule undergoing a HACA reaction under plasma conditions.

2.2. Results – negative ions

Figure 4 shows a mass spectrum of the negative ions present in a naphthalene plasma, which main feature is the detection of polyynyl anions. The main peaks of this scan are located at m/z

= 49 amu and $m/z = 73$ amu, corresponding to butadiynyl (C_4H^-) and hexatriynyl (C_6H^-) anions, respectively. In fact, a small signal corresponding to the next ion on this series (C_8H^- , octatetraynynyl anion) can be seen at $m/z = 93$ amu. The presence of these carbon chain anions in the interstellar medium was first theorised [20] and then confirmed by radio-astronomy detection techniques in 2006 [21]. This milestone triggered a systematic study of the negative ions in space, before which the existing body of knowledge on their astrophysical role was limited to H^- [22]. Since then, polyynyl anions have been detected in many sources like dark clouds and prestellar cores [23–25].

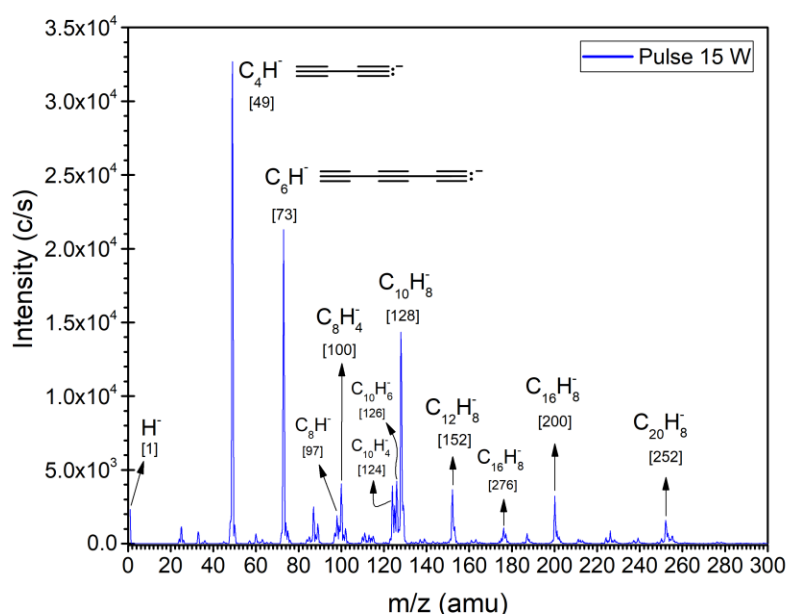


Fig. 4 - Negative ion spectrum recorded on a 15 W pulsed discharge with a 50 Hz - 50% symmetry square wave.

The formation of polyynyl anions is thought to proceed via electron attachment to the corresponding polyynes [22]. These poly-acetylenic precursors are thought to be formed through a “bottom-up” chemistry based on acetylene polymerisation [22] or addition reactions [26]. However, as an outlook to the future, it has been postulated that polyynyl anions might also be produced in a “top-down” chemistry upon destruction of PAHs [22]. In this work, we present experimental evidence that polyynyl anions can be formed as fragments of naphthalene. We postulate this mechanism based on the low concentration of acetylene in the system and the low pressure of the experiment. The occurrence of collisions between acetylene and naphthalene molecules (see previous discussion on the HACA mechanism) can be

understood only on the grounds of the great abundance of neutral naphthalene in the system. Nevertheless, we consider that collisions between 2 or even 3 acetylene molecules to yield polyynes are extremely unlikely in this system. There must be a mechanism other than acetylene polymerisation that accounts for the formation of the polyynes, precursors of the polyynyl anions C_4H^- and C_6H^- . Hence, we postulate that mechanism to be naphthalene fragmentation. Moreover, the formation of butadiyne as a naphthalene fragment was found in this experiment (see Figure 2) and supported theoretically [7]. Based on this, we believe that the polyyne which precedes the ionisation to C_4H^- is formed in a top down chemistry, as a fragment of naphthalene. Besides, the signal at $m/z = 74$ amu in the so-called set of benzene-like peaks (identified as fragments of naphthalene in Figure 2) could account for the polyacetylenic precursor needed to form C_6H^- by electron attachment.

3. R-Matrix Theory

3.1. Theoretical framework

R-Matrix theory is a framework suitable for solving the Schrodinger equation in a scattering process and thus calculating electron impact cross sectional data. For the sake of shortness, the theory will not be described in detail here although a very brief overview will be presented. The essential idea of the R-matrix method lies in dividing the space into an inner region and an outer region. In the latter, the general solution of the Schrodinger equation is either explicitly known, or can be obtained by methods which are computationally cheaper than in the inner region, where the problem must be treated in its full complexity. Once the inner region problem is solved, the R-Matrix is constructed on the on the surface dividing both regions. The R-matrix contains the scattering amplitudes at the boundaries, thus providing the boundary conditions necessary for the solution of the outer region.

These calculations are carried out by means of the freely available UKRmol+ suite of codes and with supervision from the developers of the code at Open University, Milton Keynes, England. Since it makes use of the fixed nuclei approximation, it can only calculate electronic excitation cross sections. The first step consists of a calculation of the electronic structure of the target molecule by Molpro employing HF (Hartree-Fock) when only the ground state is to be described or CASSCF (Configuration active space – self consistent field) in case electronic excitations have to be considered. This model of the molecule is the basis for the subsequent scattering calculation. The level of theory for the latter could be SE (static exchange), SEP (static exchange

plus polarisation) or CC (close coupling). In the SE and SEP models, the molecule is described with HF thus restricting the scattering channels to only the elastic one (all other channels remain closed). On the other hand, inelastic scattering is considered in CC models as the molecule can be left in an electronically excited state after the collision (one of those described by the CASSCF model).

3.2. Results and discussion

The parameter space in the target and scattering models is vast and should be explored thoroughly in order to perform meaningful scattering calculations and obtain reliable information from them. As regards the target model, we performed calculations with both a compact (cc-pVDZ) and a diffuse (6-311**G++) basis set and compared the results with theoretical and experimental data available in the literature [27,28]. We concluded that cc-pVDZ provides a better simultaneous description of the ground and excited states of the naphthalene molecule and thus decided to use this basis set thereafter for the scattering calculations. While CC calculations are still ongoing, SE and SEP calculations are completed. For these calculations, the parameters explored were the number of virtual orbitals (VOs, unoccupied orbitals of the molecule that can accommodate the scattering electron during the inner region calculation) and the number of partial waves (l , number of angular momentum components for expressing the wave function as a sum of products between the spherical harmonics and the radial wave functions).

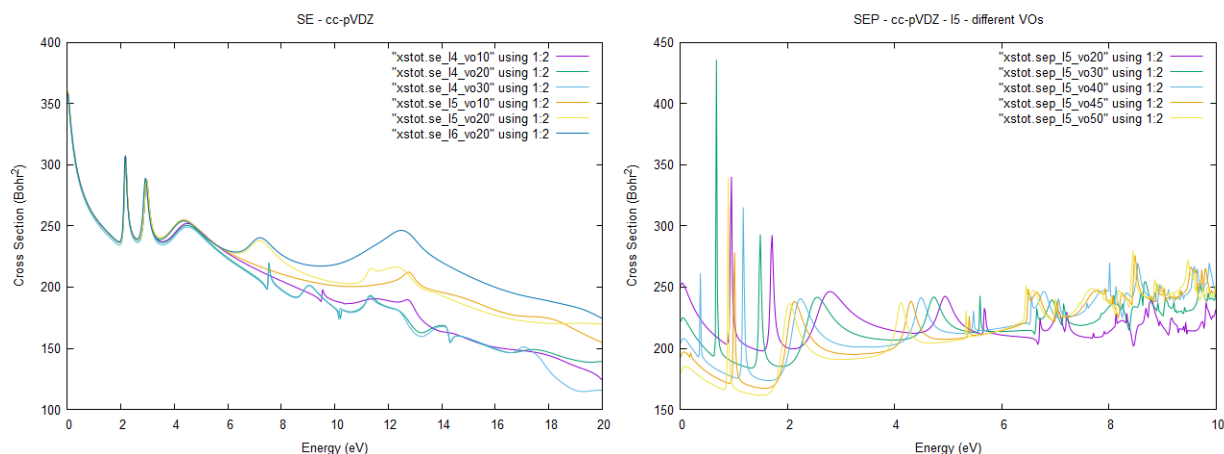


Fig. 5 – Naphthalene cross sections calculated with SE (a, left) and SEP (b, right) scattering models.

3.2.1. Static Exchange and Static Exchange plus polarisation cross sections

We performed all possible combinations between VOs=20, 30, 40, 45 and 50, and $l=4, 5, 6$, both for SE and SEP. Regarding SE, we concluded that 6 partial waves and 20 virtual orbitals was the optimum configuration, while 5 partial waves and 45 virtual orbitals was best for SEP. Some of the results are shown in Figure 5a (SE) and b (SEP).

4. Naphthalene plasma kinetic model

Building up on the zero-dimensional model for DBD reactors developed previously (End of year report 2019), a set of reactions to represent the plasma chemistry of naphthalene is under development. The chemistry of the model is based on our current understanding of the reaction pathways of naphthalene under plasma conditions, both from our experiments and from the literature. The model is being developed for naphthalene processing in a DBD reactor with Argon as carrier gas. In fact, experimental data of this mixture in a DBD reactor has been collected by another group and made available to us for validation of the model.

The model so far includes electron impact ionisation of naphthalene, fragmentation of the naphthalene radical cation, positive ion recombination, charge exchange, among other reactions. The input information for the model is retrieved from the literature when available, for instance [7] for the reaction rate coefficients of the naphthalene radical cation or [29] for the ionisation cross section. On the contrary, the elastic and electronically inelastic cross sectional data for naphthalene is being calculated via the above mentioned R-Matrix calculations. Finally, the complete set of cross sectional data for Argon is retrieved from the LXCat database.

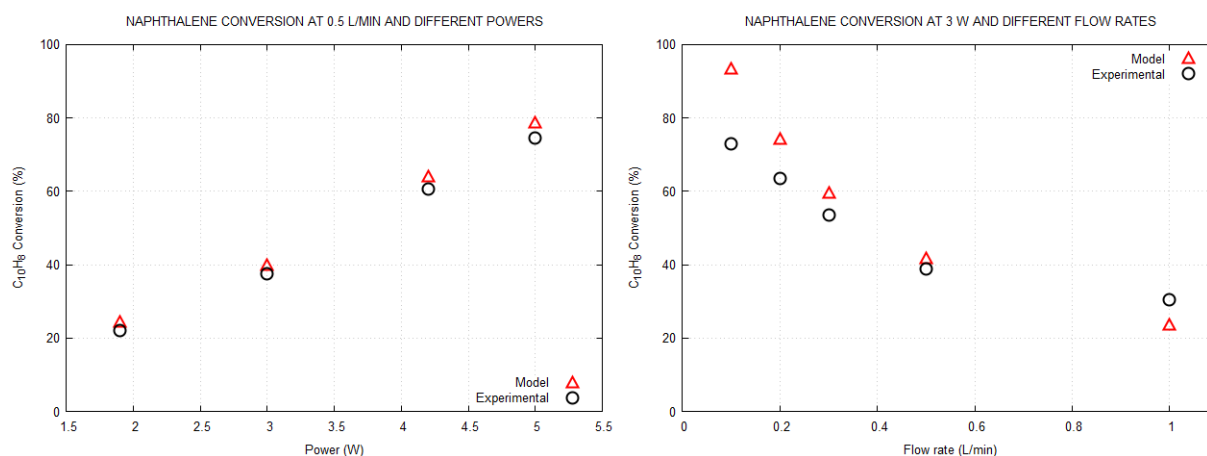


Fig. 6 – Naphthalene conversion at different powers (a, left) and flow rates (b, right). Experimental data was not produced as part of this work, rather collected elsewhere and given to us.

4.1. Preliminary results

Preliminary results of the predictions of the model compared with the experimental results with different discharge power and flow rate values are presented. Unlike the previous CO₂ model, only one fitting parameter has been used in this occasion, i.e., the frequency of pulses. This parameter has a direct correlation with the number of filaments in the discharge, and the latter, with the discharge power. This explains the good agreement with the experimental data as the discharge power varies (see Figure 6a). On the contrary, no fitting parameter has been correlated with the flowrate and thus the agreement is less satisfactory, showing high deviations towards the extremes (see Figure 6b). We believe that this can be solved by the introduction of recombination reactions (fragments going back to naphthalene), which would render a large residence time in the reactor less efficient and thus flatten the curve in Figure 6b. However, the lack of information as regards reaction rate coefficients has prevented us from further progress on this matter.

Finally, it is important to bear in mind that a reaction rate coefficient for the electron impact dissociation has not yet been included in the chemistry of the model, as it would be one of the results obtained from the R-Matrix calculations. This crucial step is what would add value to the model as it would allow for a quantitative comparison of the routes for naphthalene dissociation from which we could gain insight into the reaction mechanisms.

5. Conclusions

Regarding the experimental part of the project, the mass spectrometric study of the ionic species in a naphthalene LP-RF plasma was completed, concluding that the main dissociation channels upon electronic excitation proceed through H- or C₂H₂-loss. We reported experimental evidence of the HACA mechanism as well as the detection of polyynyl anions, i.e. C₄H⁻ and C₆H⁻. We postulate that the latter could be formed in a ‘top-down’ chemistry as fragments of PAHs. The theoretical part of the project is ongoing and shows promising results so far. It entails scattering calculations for the cross sectional data of naphthalene and plasma kinetic modelling calculations for naphthalene decomposition in a DBD reactor.

6. Output

The following papers have been published

- Plasma kinetic modelling for CO₂:
 - Alliati, M.; Mei, D.; Tu, X. Plasma Activation of CO₂ in a Dielectric Barrier Discharge : A Chemical Kinetic Model from the Microdischarge to the Reactor Scales. *J. CO2 Util.* **2018**, 27, 308–319.
- Mass spectrometry experiments with naphthalene:
 - Alliati, M.; Donaghy, D.; Tu, X.; Bradley, J. W. Ionic Species in a Naphthalene Plasma: Understanding Fragmentation Patterns and Growth of PAHs. *J. Phys. Chem. A* **2019**, 123, 2107–2113.

An abstract on the mass spectrometric naphthalene study has been submitted to ISPC24 (International symposium on plasma chemistry, Naples, Italy, June 2019). The contribution has been accepted for an oral presentation to take place on 12 June 2019.

A poster has been presented at “Explorathon”, Heriot-Watt University, Edinburgh, Scotland on 29 September 2018. This public engagement event was the Scottish contribution to the European Researchers’ night 2018.

7. Training report

The 3rd EUROPAH training has been attended. It included “PAHs, Nanoparticles and the Terrestrial Environment” at Heriot-Watt University, Edinburgh, Scotland on 25-26 September 2018; and the “Innovation circles, entrepreneurship and IP” training at the University of Liverpool on 1-2 October 2018.

The 4th EUROPAH training (“summer school: PAHs in extreme environments”) will take place at Universite Paul Sabatier, Toulouse, France on 24-28 June 2019.

8. Future plans

The targets for the upcoming period are as follows:

- R-Matrix theory (May 2019 – July 2019): complete calculations of electronically inelastic cross sections.
- Quantemol-N (July 2019 – September 2019): A secondment at Quantemol-N is planned in order to calculate the remaining naphthalene cross sections and obtain a complete set.

- Plasma kinetic modelling (May 2019 - May 2020): Complete the naphthalene model with the missing information. If required, calculate missing reaction rate coefficients through ab-initio methods. The corresponding deliverable is due by Mar 2020. Focus on other reactions thereafter.
- Output (Oct 2019 – May 2019): Several papers are expected to be published in this period, i.e., one with the R-Matrix results and a second manuscript with the plasma kinetic model of naphthalene (including the experimental data provided by the external collaborator). The DFT results on Nickel catalysts, not discussed in this report, are also expected to be published in the timeframe.

9. References

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